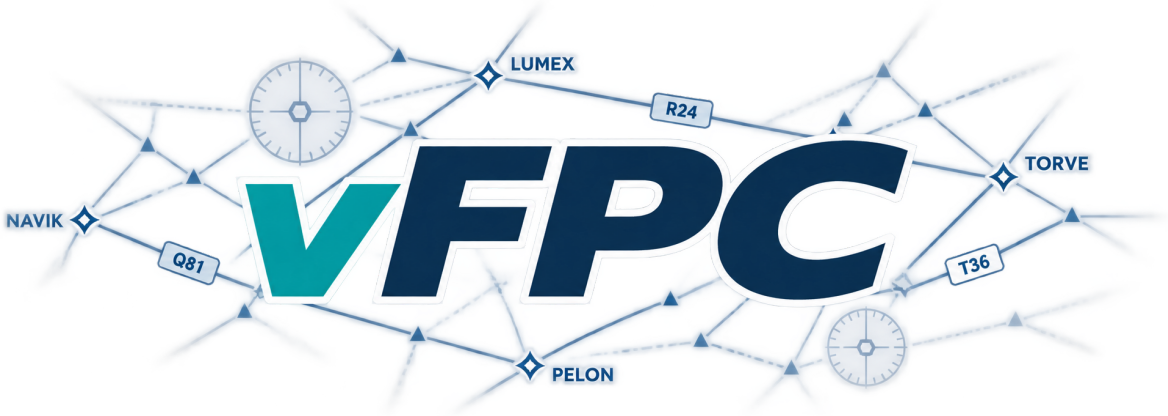


vFPC User Guide

Version 4.1.0

VATSIM Flight Plan Checker for EuroScope

Applicable to vFPC 4.1.0

VFPC Guide

vFPC is a EuroScope plugin for UK VATSIM that checks filed IFR departure flight plans against the available validation data and shows the result directly in the Departure List.

> **Note**
> Use current EuroScope UK sector files and the VATSIM UK Controller Pack before using vFPC.

Updates in 4.1.0

Sector controllers can use vFPC across multiple departure airports. Recognised APP and CTR positions validate departure-active Runway Activity airports within their configured UKCP coverage. See Controller Scope for setup, exclusions and lookup-failure behaviour. This support concerns departure flight plans rather than validation of aircraft progress through a sector.

Exit-point diagnostics distinguish known prohibited exits from unidentified exits, and warn when a permitted exit is extracted but not explicitly filed. Existing validation failures retain priority over the warning. See Exit Point Diagnostics and Warnings for the SUMMARY and FULL messages.

The update also corrects route suggestions so they respect applicable constraints and SID restrictions, preserves OFF reporting for out-of-scope flights, and improves pending-result handling and support logging. These fixes are included alongside the sector-controller and exit-point features.

Guide Version

This guide applies to **vFPC 4.1.0**. It covers sector-controller support, exit-point diagnostics and warnings, and the fixes described above.

Guides

- Installation and Configuration - installing and configuring vFPC.
- Using vFPC - normal operation, sector-controller setup and validation results.
- Validation Reference - result codes and how to interpret them.
- Troubleshooting and Limitations - common problems and known limitations.
- Reporting an Issue - getting support and reporting issues.

> **Controller Responsibility**
> vFPC is a controller aid, not a clearance engine. Validation results do not replace current procedures, local instructions or controller judgement.

Installation and Configuration

This guide explains how to install the vFPC plugin into EuroScope and configure it for use. The process only needs to be completed once.

Download

vFPC is no longer included with the VATSIM UK Controller Pack.

Download the latest version from the GitHub Releases page.

After downloading `VFPC.dll` move it to the folder `%APPDATA%\EuroScope\UK\Data\Plugin\VFPC\`.

The plugin will work from any location, but the above path is the recommended location for consistency with other EuroScope plugins.

> **Note**
> vFPC stores its configuration and data files using the standard Windows Application Data folders. These locations are determined automatically by Windows and no configuration is normally required. For more information about **%APPDATA%** and **%LOCALAPPDATA%** folders see the Microsoft documentation: - Microsoft Learn - KnownFolder IDs

Plugin Configuration

Before you use the plugin it needs to be configured within EuroScope as follows:

> **Allow Draw on Types**
> The plugin does not require **Allowed To Draw On Types** to function. Please **DO NOT** alter these settings.

Load the plugin

1. Start EuroScope and open the required profile (`.prf`).
2. From the menu bar, select **Other Set** → **Plug-ins**.
3. From the Plug-ins Dialog, click the **Load** button and navigate to the folder where you moved `VFPC.dll`.
4. Select `VFPC.dll` and click **Open**.
5. `VFPC.dll` should now appear in the list as shown below.

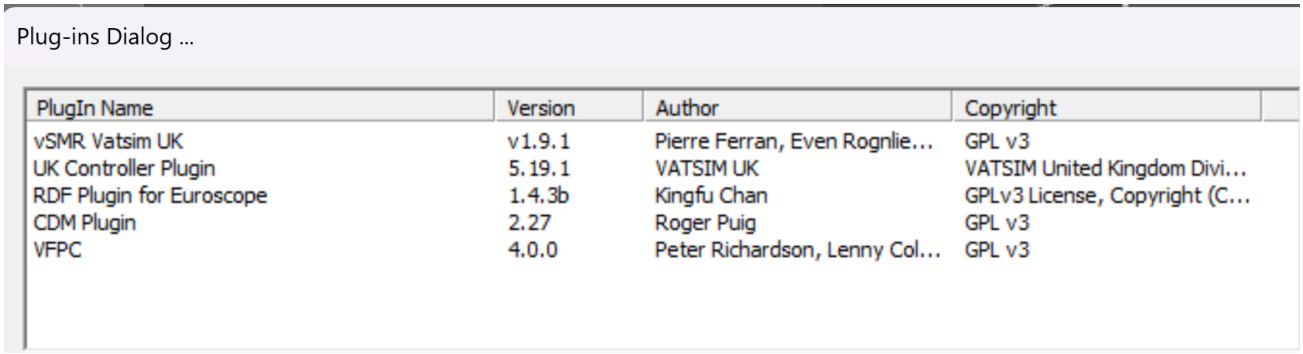


Figure 1: EuroScope Plugin Dialog List.

Departure List Configuration

The final part of configuration is to add a column to the Departure List which is used to display the validation results.

Add the vFPC column

1. Click **S** in the top left of your departure list to open the 'Departure list columns setup dialog'.

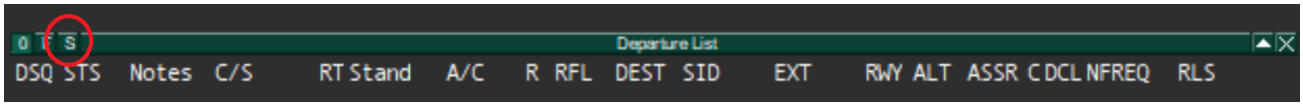


Figure 2: Departure list menu.

2. Click **Add Item**. A new item named **Callsign** is added to the bottom of the list.
3. Select the new item and change **Tag Item Type** to **VFPC/VFPC**.
4. Set **Header name** to **FPL**, **Width** to **4**, and leave **Align to center** unchecked.

Configure mouse actions

1. Set **Left button** to **VFPC / Options**.
2. Set **Right button** to **UK Controller Plugin / Open SRD Search**.
3. The Departure List columns setup dialog should now look similar to Figure 3.

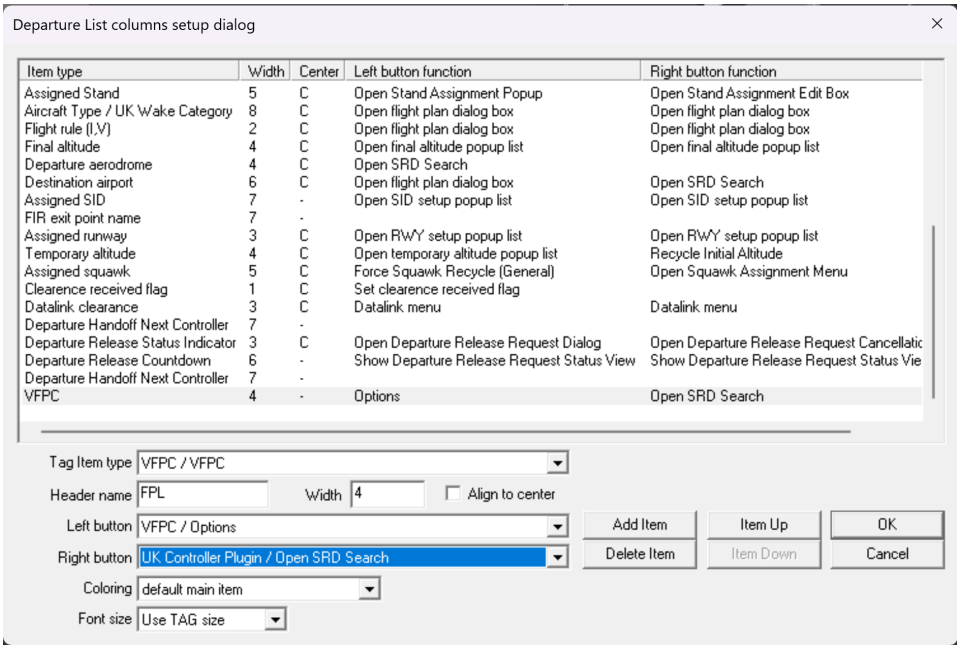


Figure 3: Departure list columns setup

The vFPC plugin is now configured and ready for use.

> **FPL Column Position**
> You can move the position that the FPL column appears on the Departure List by using the 'Item Up' and 'Item Down' buttons.

Updating vFPC

New versions of vFPC are published on the GitHub Releases page.

To update vFPC:

1. Close EuroScope.
2. Download the latest version of `VFPC.dll`.
3. Replace the existing `VFPC.dll` in your vFPC plugin folder with the new version.
4. Restart EuroScope.

Your existing Departure List configuration does not normally need to be changed when updating vFPC.

> **Note**
> Release notes are provided with each release and should be reviewed for any additional installation or configuration requirements.

Using vFPC

Once installed and configured, vFPC operates automatically while you are controlling. Validation results are displayed in the **FPL** column of the EuroScope Departure List.

Normal Operation

vFPC validates IFR departure flight plans for the airports selected within the current controller scope, using each flight's origin-airport data.

A flight plan is normally validated when:

- A new IFR flight plan is detected.
- A relevant part of an existing flight plan is changed.
- Updated validation data is loaded.

VFR flight plans are not validated.

> **Note**
> vFPC requires a departure runway to be assigned before flight plan validation can take place.

Controller Scope

Using vFPC as an APP or CTR Sector Controller

vFPC supports sector controllers reviewing departures from multiple airports. Recognised **APP** and **CTR** callsigns can validate IFR departures from the departure-active Runway Activity airports that fall within the position's configured UKCP coverage. You do not need to operate a separate airport-specific position for each departure airport.

1. Connect using your recognised full controller callsign, including any sector identifier, for example `SCO_CTR` or `SCO_SOL_CTR`.
2. In EuroScope **Runway Activity**, enable the airport-level **DEP** selection for each departure airport you want vFPC to review.
3. Ensure each airport is covered by that position's UKCP mapping. vFPC excludes selected airports outside that coverage and explains the exclusion.
4. Ensure the individual flight has an assigned departure runway. Each eligible flight is checked using its own origin airport's validation data.

The airport-level DEP selection matters: selecting runway departure symbols alone, or making an arrival-only selection, does not select the airport for sector-controller departure validation. An airport appearing in the Departure List does not automatically put it in vFPC scope.

For example, `SCO_SOL_CTR` can review selected departures from EGPH, EGPF and EGPK when those airports are within its configured coverage. Selecting EGLL does not add it to that sector's scope. An out-of-scope flight displays **OFF**; an eligible flight awaiting airport data or its own departure runway displays **PDG**.

Other Positions and Scope Rules

vFPC matches the full controller callsign against UKCP controller-position data. For a recognised airport-specific DEL, GND or TWR position, it selects the mapped airport even when that airport is not departure-active in Runway Activity. A warning explains the missing Runway Activity selection; it does not remove the airport from scope.

For a callsign ending in `_OBS`, departure-active Runway Activity airports define the review scope without a UKCP filter. No airport is inferred from aircraft in the Departure List. UKCP coverage describes configured position coverage; it does not determine which other controllers are online or resolve delegation.

Unrecognised operational positions, unsupported suffixes and empty UKCP coverage pause validation. Flights outside the selected scope display `OFF`. In-scope flights still wait for airport data and their own assigned departure runway; these pending states display `PDG`.

Support Commands - vFPC Team Advice Only

These commands are for troubleshooting only. Use them only when advised by the vFPC team; they are not required for normal setup or operation.

- `.vfpc scope` displays the selected airports, selection explanation and UKCP lookup status.
- `.vfpc scope reload` requests fresh UKCP position data. It does not change the airport constraint source or EuroScope Runway Activity settings.

UKCP data is held in memory for the connected session and loaded on connection. A failed refresh retains valid session data. With no usable lookup, operational validation pauses; OBS selection remains available. There is no automatic retry loop. If a lookup failure persists, contact the vFPC team for advice. Disconnecting clears the scope cache. A failure to read Runway Activity pauses APP/CTR/OBS selection; saving the Runway Activity dialog requests a new scan. Fixed local scope remains eligible.

The FPL Column

The **FPL** column within the Departure List displays a short code indicating the current validation result.

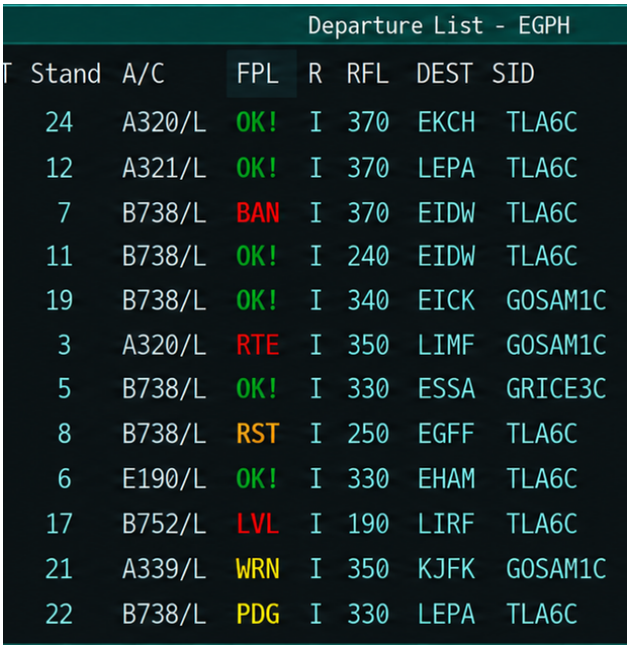


Figure 4: vFPC validation results in the EuroScope Departure List.

For example:

- **OK!** — the flight plan passed validation.
- **WRN** — advisory information is available.
- **RTE** — the route requires attention.
- **PDG** — validation is pending.
- **VFR** — the flight plan is VFR and is not validated.
- **OFF** — the flight is outside the selected controller scope, checks are disabled, or EuroScope is disconnected.
- **OVR** — a controller override has been applied.

The code is intended to provide a quick indication only. See the Validation Reference for a full explanation of each result.

Using the FPL Column

The **FPL** column is also used to access vFPC functions.

Left-click — vFPC Options

Left-click the **FPL** result for a flight plan to open the vFPC options menu.

As shown in **Figure 5**, the menu provides four options:

- **Show Checks (SUMMARY)**
- **Show Checks (FULL)**
- **Toggle Override**
- **Toggle Checks**

The purpose and use of each option are described in the following sections.

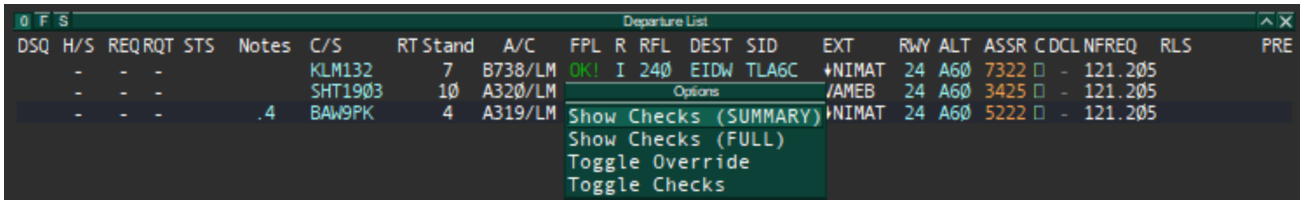


Figure 5: vFPC options available from the Departure List.

Right-click — SRD Search

Right-click the **FPL** result to open **SRD Search** using the UK Controller Plugin.

This provides quick access to the applicable Standard Route Document information when reviewing a flight plan.

Viewing Validation Results

More detailed information about a validation result can be displayed in the EuroScope message window by selecting **Show Checks (SUMMARY)** or **Show Checks (FULL)** from the menu shown in **Figure 5**.

Show Checks (SUMMARY)

SUMMARY displays the principal validation result together with a short explanation. This will normally provide sufficient information to understand why a flight plan has not validated successfully.

Show Checks (FULL)

FULL displays the individual checks performed by vFPC and their results. Use this when more detail is required or when investigating an unexpected validation result.

> **Note**
> Validation details are displayed in the EuroScope message window only when **Show Checks (SUMMARY)** or **Show Checks (FULL)** is selected.

Exit Point Diagnostics and Warnings

vFPC distinguishes the route explicitly filed in the flight plan from EuroScope's extracted waypoint list. The extracted list can include intermediate waypoints that are not written individually in the filed route, for example along an airway.

Exit-point validation uses the extracted waypoint list and the departure airport's constraints. Route validation remains a separate check of the normalized filed route. Finding an exit in the extracted list does not by itself establish that the point is permitted for the filed departure or that all other route requirements have passed.

XIT for an Exit Point Failure

When a known exit point is not permitted for the filed departure, **SUMMARY** identifies the point and departure. For example:

```
ALOTI not permitted via BPK
```

FULL explains the failure and lists other departures only when supported by the airport's exit-point constraints. For example:

> ALOTI is not a permitted exit point for the filed BPK departure. ALOTI is permitted via UMLAT or ULTIB departures.

The alternative departures are an unordered list; their order does not indicate preference or priority. Exit-point advice does not establish that the complete flight plan would pass every other check via an alternative departure. Review the applicable route and procedures before making a change.

If no known exit is identified, **SUMMARY** displays:

> No known exit point identified

FULL explains that no known exit was identified in the extracted waypoint list for the filed departure. This differs from a known point that is not permitted for that departure. Later checks may be reported as not checked because exit-point validation failed.

WRN for a Permitted Exit Missing from the Filed Route

When an exit is permitted for the filed departure and appears in the extracted waypoint list but is not explicitly included in the filed route, the exit-point check passes and vFPC records a warning. If there is no higher-priority failure, the FPL column displays **WRN**.

Example **SUMMARY**:

> ALOTI absent from filed route

Example **FULL** warning:

> ALOTI is a permitted exit point for the filed departure and appears in EuroScope's extracted waypoint list, but is not explicitly included in the filed route.

This warning is not an XIT failure. Review the filed route and EuroScope's interpretation rather than treating the warning alone as proof of an invalid route. The actual identified exit point is used in the message.

A Failure Takes Priority over WRN

An existing validation failure retains its FPL code when the missing-filed-exit warning also applies. For example, **OER** remains displayed for an odd/even level failure even if a permitted exit is absent from the filed route.

Select **Show Checks (FULL)** and review **Warnings** to see the warning alongside the failure. SUMMARY normally explains the principal failure rather than displaying every warning. No additional unsolicited console messages are generated for these exit diagnostics; use Show Checks to request the details.

Toggle Override

Toggle Override allows the controller to override the validation status displayed in the **FPL** column.

An override does not change the filed flight plan or the underlying validation result. The **FPL** column displays **OVR** to indicate that a controller override has been applied.

Select **Toggle Override** again to remove the override.

Toggle Checks

Toggle Checks disables validation for the selected flight plan. The **FPL** column displays **OFF** while validation is disabled.

Select **Toggle Checks** again to re-enable validation.

This can be useful where vFPC validation is not appropriate for a particular flight plan.

When a Flight Plan Does Not Validate

A result other than **OK!** does not necessarily mean that the flight plan should simply be rejected or changed.

1. Review the code displayed in the **FPL** column.
2. Select **Show Checks (SUMMARY)** to view the reason.
3. If further information is required, select **Show Checks (FULL)**.
4. Compare the result with the applicable procedures and current flight plan.
5. Take the appropriate controller action.

If vFPC does not behave as expected, see Troubleshooting and Limitations.

> **Controller Responsibility**
> vFPC is a controller aid. Validation results should be considered alongside current procedures and the controller's own assessment of the flight plan.

Validation Reference

vFPC displays a short validation code in the **FPL** column of the EuroScope Departure List. The code provides a quick indication of the validation status of the flight plan.

This page explains the validation colours and result codes used by vFPC. For information about viewing Summary and Full validation results, see Using vFPC.

How Validation Works

When an IFR flight plan is ready for validation, vFPC checks the filed information against the data available for the departure airport.

The diagram below provides a simplified overview of the validation process. Individual checks may depend on the results of earlier checks.

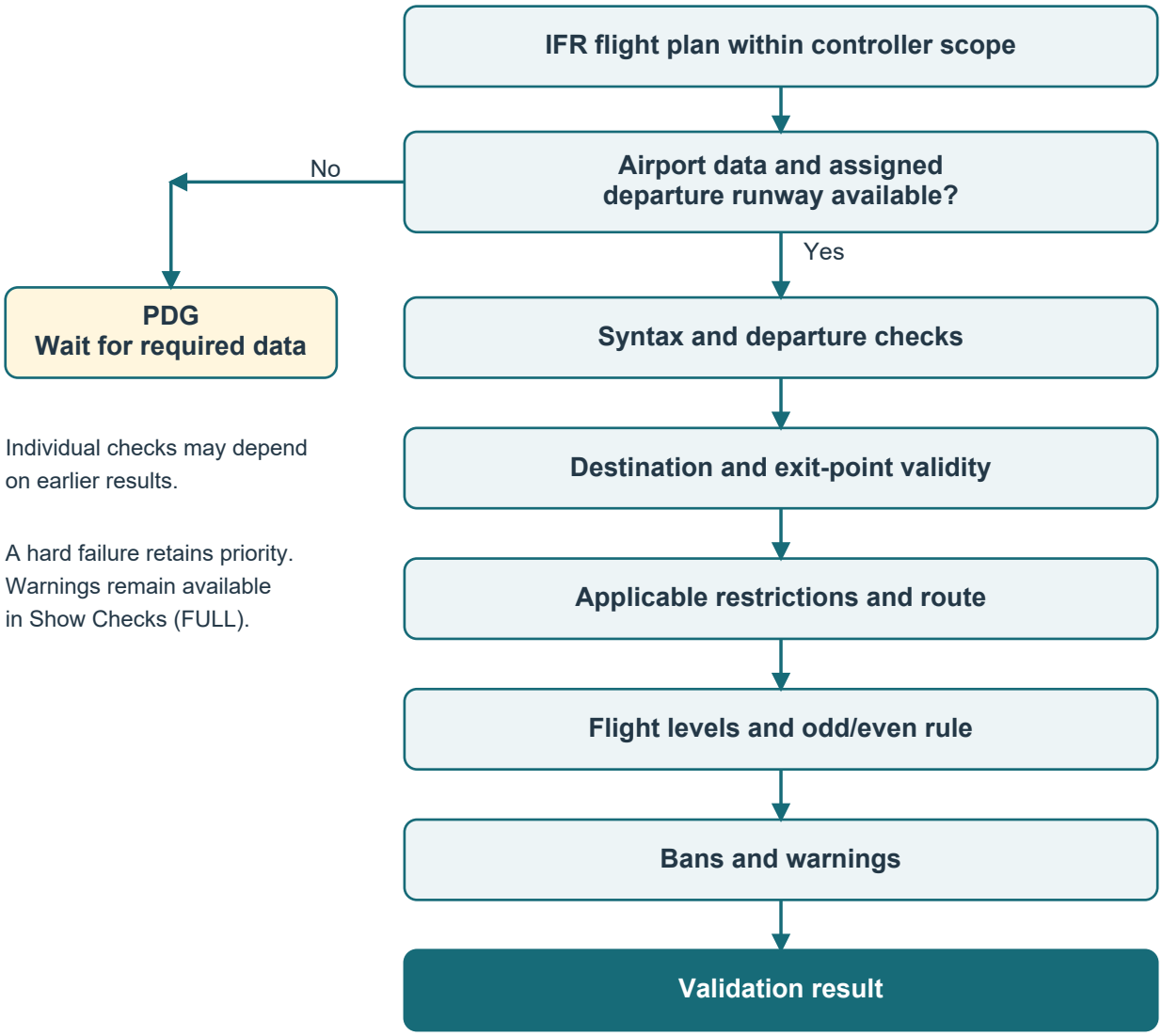


Figure 6: Simplified vFPC validation process.

The validation data is primarily generated from the NATS Standard Route Document (SRD) and may also include airport-specific operational rules and local procedures.

Result Colours

The colour of the FPL column provides the quickest indication of the validation status.

Colour	Meaning
Green	Flight plan validated successfully.
Yellow	Validation is pending or advisory information is available.
Amber	Flight plan contains a level, odd/even or suffix issue that should be reviewed.
Red	Validation failure. Controller action required.
Grey	Flight plan is not currently being validated (for example VFR traffic or validation disabled).
Cyan	Controller override has been applied.

> **Note**
> The exact colours used are configurable within the vFPC configuration file. The colours shown above represent the default behaviour.

Result Codes

The following table provides a quick reference to the result codes displayed in the **FPL** column. More detailed information about each result is provided below.

Code	Meaning
OK!	All validation checks passed
WRN	Advisory information is available
PDG	Validation is pending
VFR	VFR flight plan — not validated
OFF	Outside selected controller scope, checks disabled, or EuroScope disconnected
OVR	Controller override applied
SID	SID validation failed
DST	Destination validation failed
XIT	Exit-point validation failed
RTE	Route validation failed
LVL	Flight-level validation failed
OER	Odd/even flight level validation failed
SFX	SID suffix validation failed
RST / **RST**	SID or route restriction requires attention
BAN	Applicable route or SID ban triggered
SYN	Flight plan syntax validation failed
ERR	Error encountered during validation

OK! indicates that all applicable validation checks have passed.

No further vFPC action is required.

WRN indicates that the flight plan has passed validation, but advisory information is available.

Review the Summary result for details of the warning. One warning condition is a permitted exit point that appears in EuroScope's extracted waypoint list but is absent from the filed route. The exit-point check passes in this case.

If another check fails, its failure code takes priority over WRN. The warning remains available under **Warnings** in **Show Checks (FULL)**. See Exit Point Diagnostics and Warnings for examples.

PDG indicates that validation is pending.

This is normally a temporary condition while vFPC is waiting for information required to validate the flight plan, for example:

- Airport validation data is still loading.
- A departure runway has not yet been assigned.

Validation will proceed when the required information becomes available.

VFR indicates that the flight plan is VFR.

vFPC validates IFR departure flight plans only. VFR flight plans are therefore not validated.

OFF indicates that the flight is outside the selected controller scope, checks have been disabled using **Toggle Checks**, or EuroScope is disconnected.

For a scope exclusion, use **Show Checks (SUMMARY)** or **Show Checks (FULL)** to read the origin/scope explanation. Only when advised by the vFPC team, use `.vfpc scope` to inspect the selected airports. Selecting **Toggle Checks** does not bring an excluded airport into scope.

If checks were explicitly disabled, select **Toggle Checks** again to re-enable them. An in-scope VFR flight is reported as VFR rather than OFF.

OVR indicates that the controller has applied an override to the validation status displayed in the **FPL** column using **Toggle Override**.

The override does not change the filed flight plan or the underlying validation result.

Select **Toggle Override** again to remove the override.

SID indicates that the flight plan has failed SID validation.

Review the Summary result for details of the SID validation failure.

DST indicates that the destination has failed validation against the applicable route requirements.

Review the Summary result for details of the destination validation failure.

XIT indicates that the exit point has failed validation against the applicable route requirements.

SUMMARY distinguishes a known exit point that is not permitted for the filed departure from a case where no known exit point was identified. FULL may list constraint-supported alternative departures; their order does not imply preference. No alternative is listed when the data does not support one.

A permitted exit that is extracted but absent from the filed route is a warning, not an XIT failure. See Exit Point Diagnostics and Warnings for the messages and how to review them.

RTE indicates that the filed route has failed validation against the applicable route requirements.

Review the Summary result for details of the route validation failure.

Suggested routes and mismatch explanations are limited to constraints that survived the earlier checks and meet the requested level and odd/even rules. Suggestions also respect applicable SID restrictions, including aircraft type,

suffix, EOBT and prohibitions. Duplicate alternatives are omitted; vFPC does not widen the suggestion set to routes already rejected by those checks.

These suggestions help explain an RTE result. They do not change the filed flight plan or replace the controller's assessment of the complete route and current procedures.

LVL indicates that the requested flight level has failed validation against the requirements for the applicable route.

Review the Summary result for details of the flight level validation failure.

OER indicates that the requested flight level has failed the odd/even flight level requirements for the applicable route.

Review the Summary result for details of the odd/even validation failure.

SFX indicates that the SID suffix has failed validation against the requirements for the applicable route.

Review the Summary result for details of the SID suffix validation failure.

RST or **RST** indicates that a SID or route restriction applies to the flight plan.

- **Red** indicates that the restriction has caused validation to fail.
- **Amber** indicates that the restriction is advisory.

Review the Summary result for details of the applicable restriction.

BAN indicates that the flight plan has triggered an applicable route or SID ban.

Review the Summary result for details of the applicable ban.

SYN indicates that the flight plan has failed syntax validation.

Review the Summary result for details of the syntax validation failure.

ERR indicates that vFPC encountered an error while attempting to validate the flight plan.

Review the Summary result for further information. If the cause is not clear, refer to Troubleshooting and Limitations.

> **Controller Responsibility**
> vFPC is an advisory validation tool designed to assist controllers. Controllers remain responsible for applying current procedures, local instructions, SRD requirements, SID restrictions, level capping requirements and the semicircular rule. An **OK!** result should not be used as a substitute for operational knowledge and judgement.

Troubleshooting and Limitations

This page covers common issues that may prevent vFPC from validating a flight plan or displaying the expected result.

For information about individual validation result codes, see the Validation Reference.

vFPC Does Not Appear in the Departure List

Check that:

- `VFPC.dll` has been loaded in EuroScope.
- The **VFPC/VFPC** tag item has been added to the Departure List.
- The **FPL** column is visible.

If necessary, repeat the configuration steps in Installation and Configuration.

A Flight Plan Is Not Being Validated

vFPC validates IFR departure flight plans for airports within the current controller scope.

A flight plan may not be validated if:

- The flight plan is VFR.
- A departure runway has not yet been assigned.
- Validation checks have been disabled using **Toggle Checks**.
- The departure airport is outside the current controller scope.
- For APP/CTR, the airport is not departure-active in Runway Activity or is outside the position's configured UKCP coverage.
- The required airport validation data is not available.

The code displayed in the **FPL** column may provide more information. See the Validation Reference.

Validation Remains Pending

A **PDG** result indicates that vFPC is waiting for information required to perform validation.

This may occur while airport validation data is loading or when a departure runway has not yet been assigned.

If **PDG** remains displayed after the required information is available, select **Show Checks (SUMMARY)** or **Show Checks (FULL)** and review the information displayed in the EuroScope message area.

An Unexpected Validation Result Is Displayed

If a result does not appear to agree with the filed flight plan:

1. Select **Show Checks (SUMMARY)**.
2. If necessary, select **Show Checks (FULL)** for the individual validation checks.
3. Compare the result with the applicable UK procedures and the filed flight plan.

If the result still appears incorrect, see Reporting an Issue.

An Exit Warning Is Not Shown in the FPL Code

An existing validation failure takes priority over **WRN**. Select **Show Checks (FULL)** and review **Warnings**. A permitted exit can be absent from the filed route while the FPL code reports a separate failure, such as **OER**.

An Exit Is Missing from the Filed Route

EuroScope can extract a permitted intermediate waypoint that is not explicitly filed. In that case vFPC passes the exit-point check and records a warning. Review the filed route and the extracted waypoint list; do not interpret the warning alone as an XIT failure.

If no known exit is identified in the extracted list, vFPC instead reports **XIT**. Deleting a waypoint from the filed route does not guarantee that it will remain in EuroScope's extracted list. See Exit Point Diagnostics and Warnings.

Known Issues

There are currently no known issues.

Issues that have been resolved are documented in the release notes.

Known Limitations

vFPC is designed to validate IFR departure flight plans against the available vFPC validation data. It does not validate VFR flight plans.

Validation is dependent on the coverage and accuracy of the available validation data. A successful validation does not remove the controller's responsibility to ensure that a flight plan and clearance are appropriate.

Reporting an Issue

If you are unable to resolve a problem using the Troubleshooting and Limitations page, support is available through the vFPC Discord server.

Getting Support

Joining the vFPC Discord Server

Use the link below to join the vFPC Discord server:

Join the vFPC Discord server

If you do not already have a Discord account, you will be prompted to create one or sign in before joining the server.

Raising a Support Ticket

Once you have joined the server, open a support ticket in the appropriate support channel.

Information to Provide

When reporting a problem, provide as much of the following information as applicable:

- vFPC version.
- EuroScope version.
- Callsign.
- Departure and destination aerodromes.
- Assigned departure runway.
- SID.
- EOBT.
- Aircraft type.
- Requested flight level.
- Filed route.
- The result displayed in the **FPL** column.
- The **SUMMARY** result.
- The **FULL** result, where relevant.
- What result you expected.
- The relevant vFPC log file.
- The startup log entry showing the version and build identification.

Using the vFPC Log File

vFPC creates a new log file each time the plugin is started. The log contains diagnostic information about the operation of vFPC and flight plan validation.

The log files are normally located in:

`%APPDATA%\EuroScope\UK\Data\Plugin\VFPC\log`

Log files use the following naming format:

`vfpc-YYYYMMDDHHMMSS.log`

The log file can be useful when investigating an unexpected validation result or other problem with vFPC.

When reporting a problem, you may be asked to attach the log file covering the period when the problem occurred.

Build and Validation Information in Logs

The startup entry identifies the plugin version, numeric DLL revision, build configuration and source/build identity. Include that entry when reporting a problem so support can distinguish the build being used.

Logs also record validation results when airport data becomes available, not only when the flight is first seen. Airport-data load and lifecycle information helps support investigate loading or shutdown problems. You do not need to interpret the memory or source-identity fields yourself; provide the relevant log when requested.

Detailed Logging

vFPC supports additional diagnostic logging for investigating problems.

Detailed logging should normally only be enabled when requested by vFPC support. Instructions for enabling it will be provided if required.